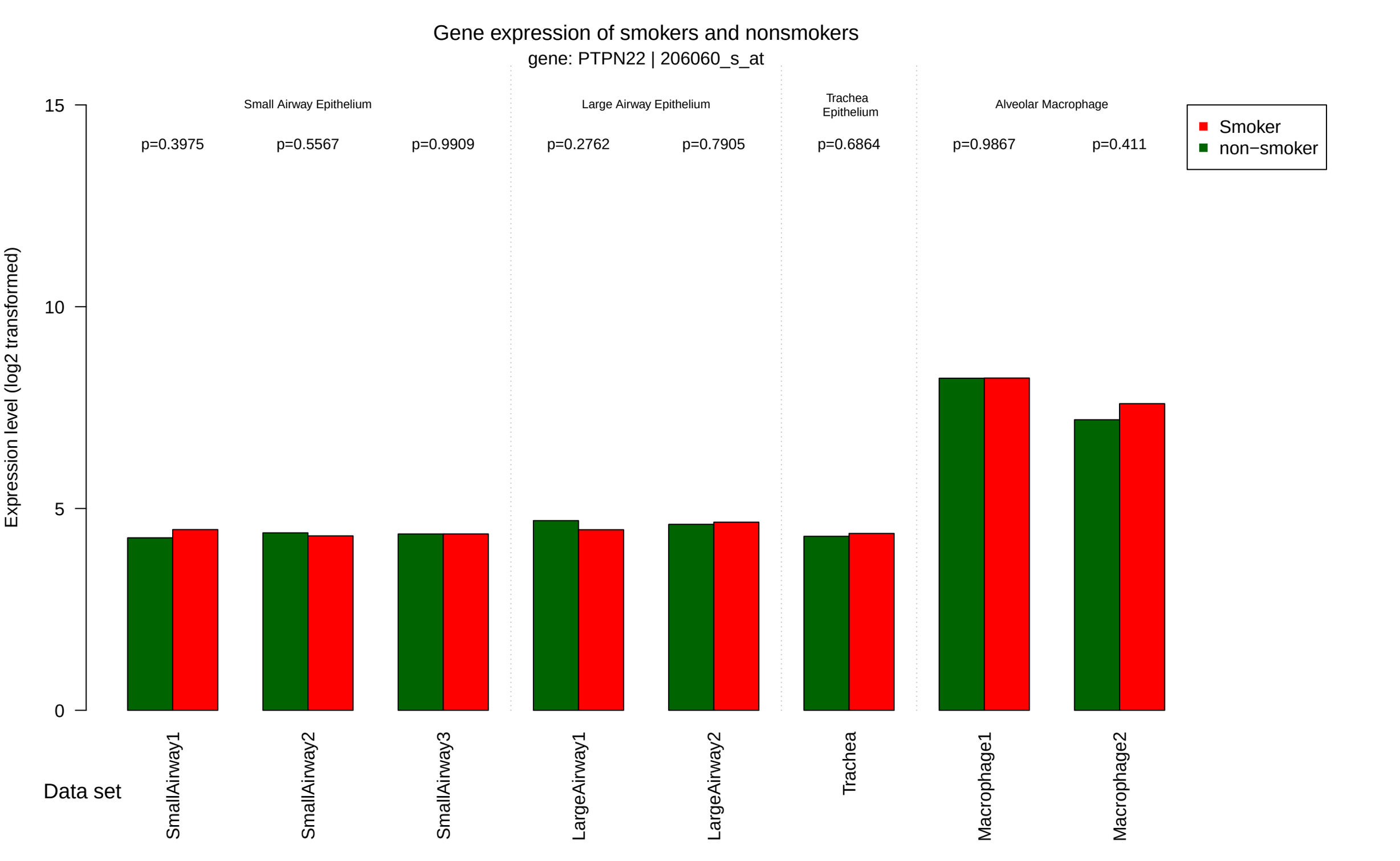
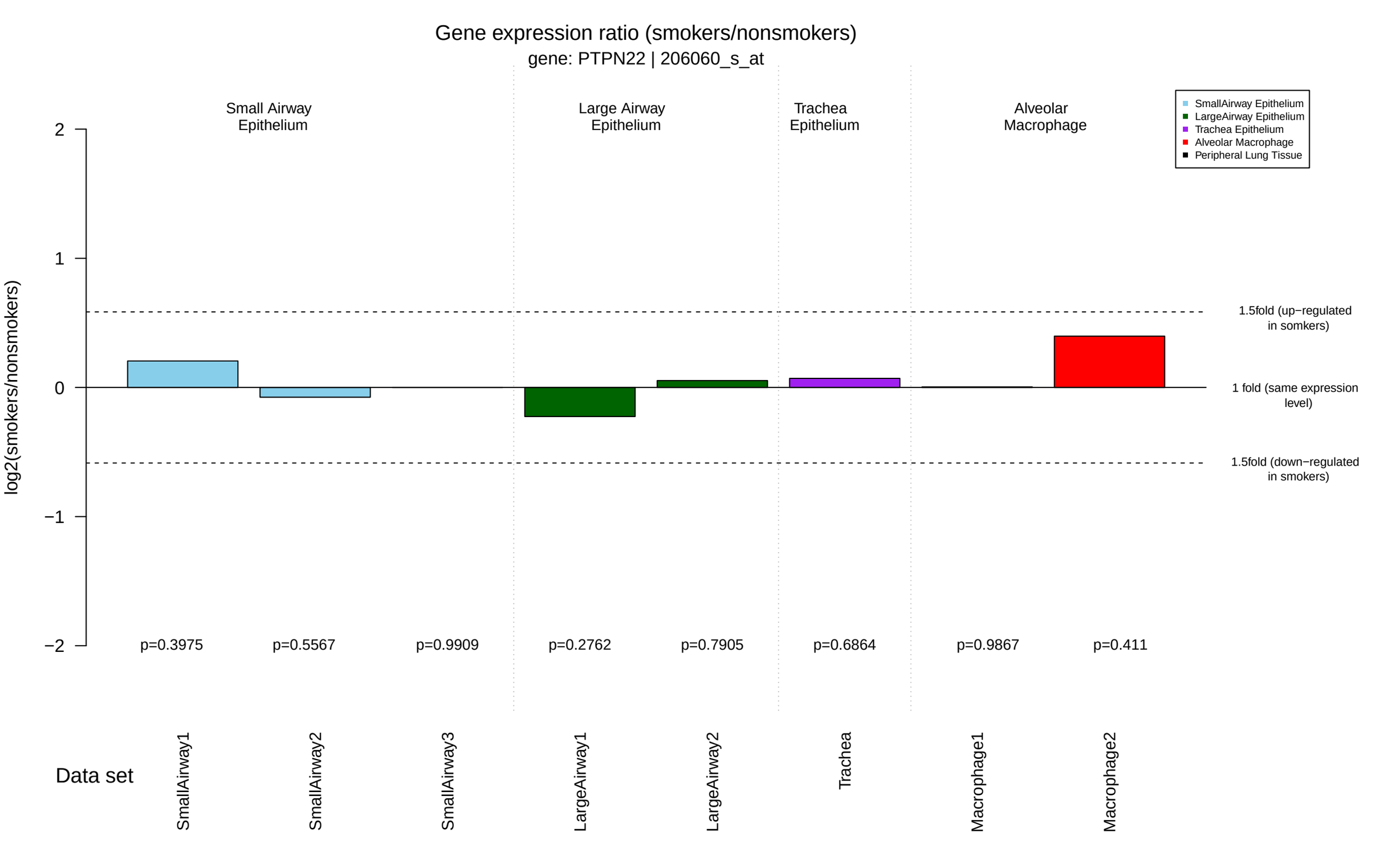
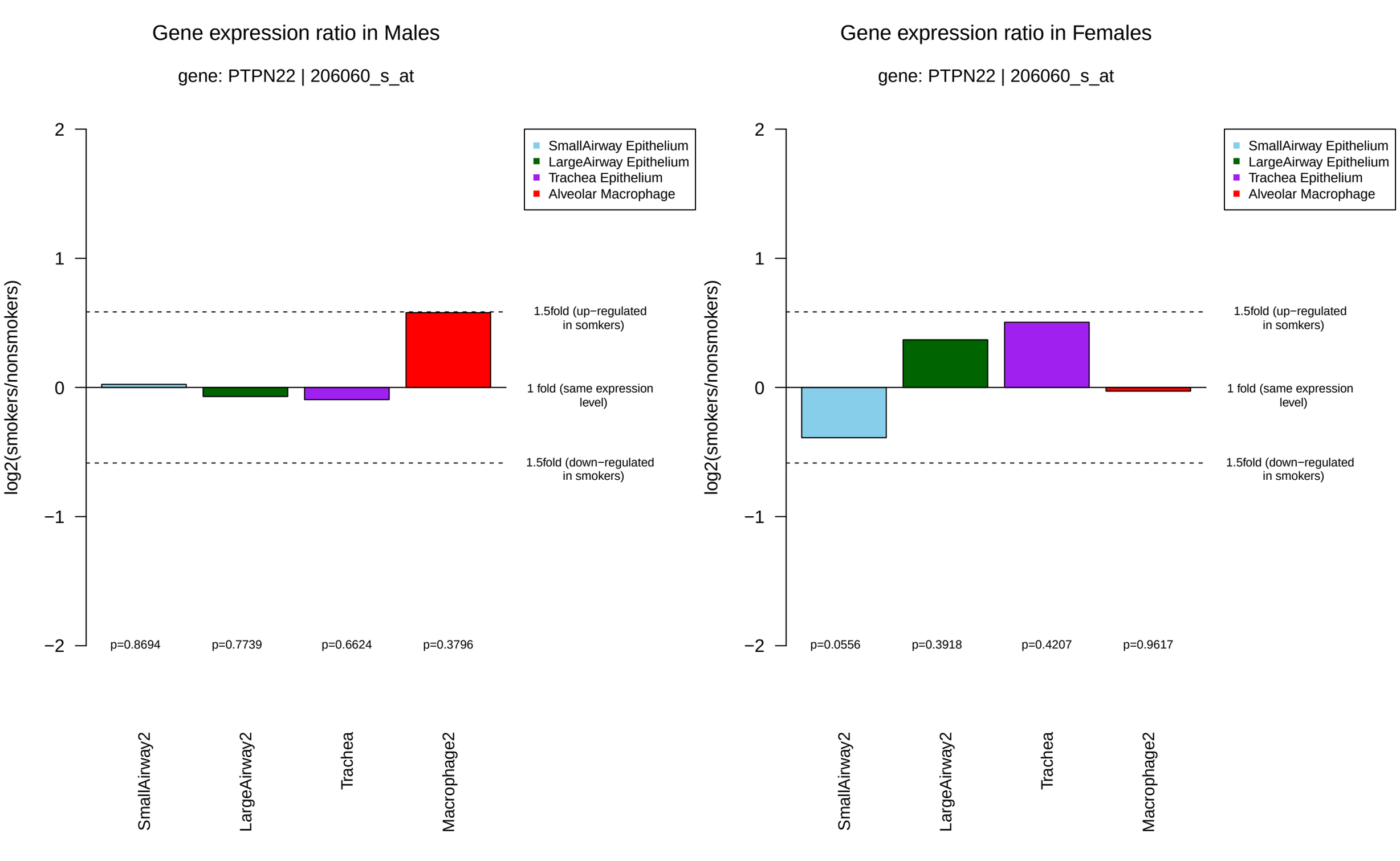
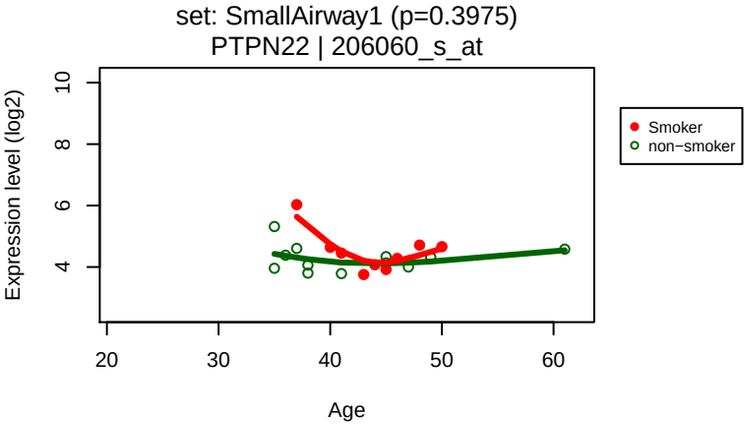
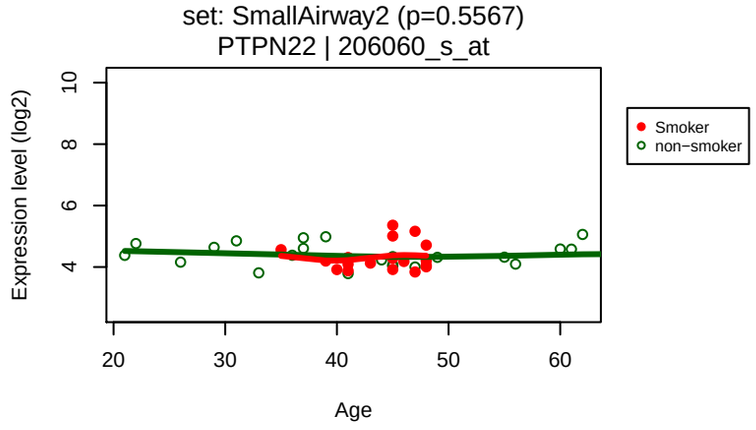
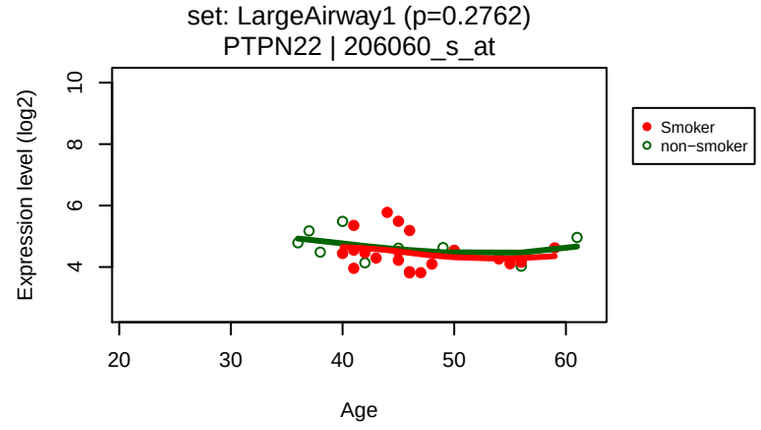

Gene expression of smokers and nonsmokers
gene: PTPN22 | 206060_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3975
p=0.5567
p=0.9909
p=0.2762
p=0.7905
p=0.6864
p=0.9867
p=0.411
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

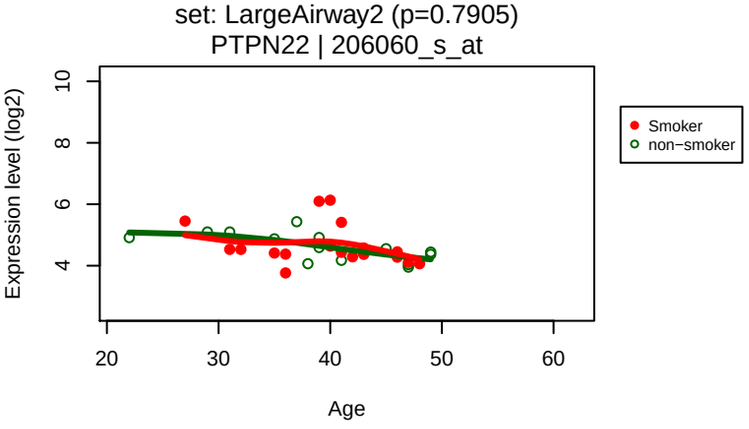

Gene expression ratio (smokers/nonsmokers)
gene: PTPN22 | 206060_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3975
p=0.5567
p=0.9909
p=0.2762
p=0.7905
p=0.6864
p=0.9867
p=0.411
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

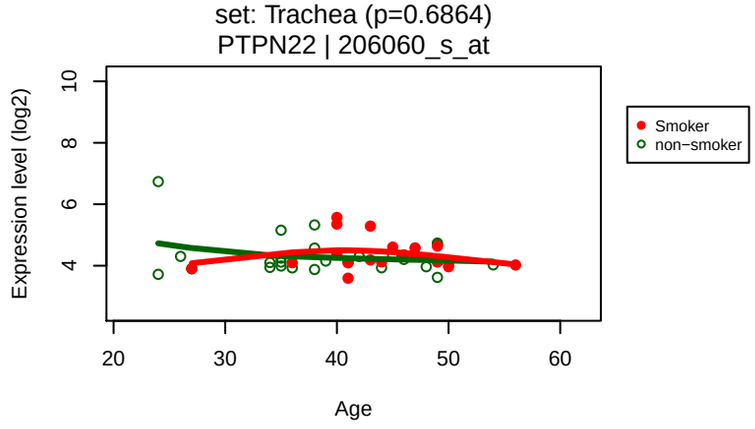

Gene expression ratio in Males
gene: PTPN22 | 206060_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8694
p=0.7739
p=0.6624
p=0.3796
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PTPN22 | 206060_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0556
p=0.3918
p=0.4207
p=0.9617
SmallAirway2
LargeAirway2
Trachea
Macrophage2

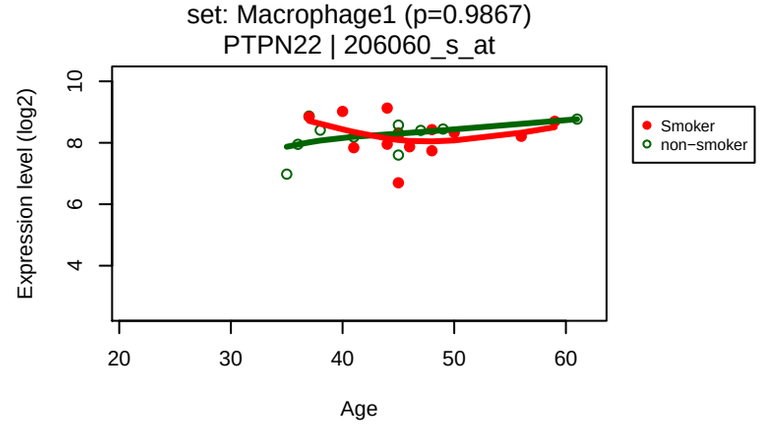

set: SmallAirway1 (p=0.3975)
PTPN22 | 206060_s_at
Expression level (log2)
Age
Smoker
non−smoker

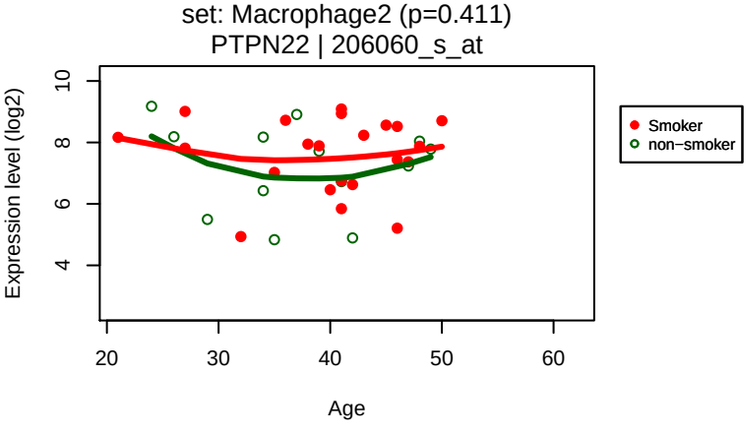

set: SmallAirway2 (p=0.5567)
PTPN22 | 206060_s_at
Expression level (log2)
Age
Smoker
non−smoker

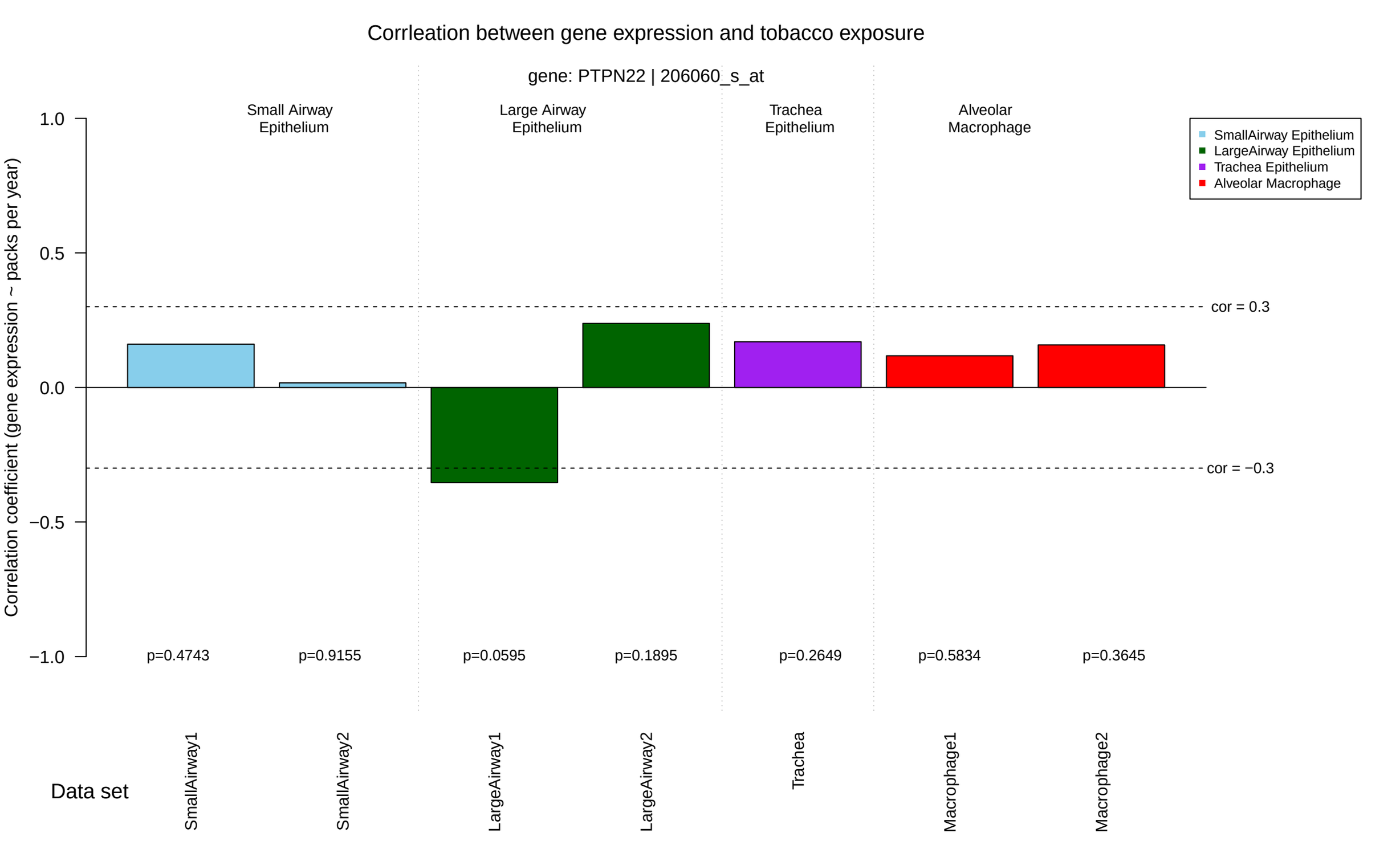

set: LargeAirway1 (p=0.2762)
PTPN22 | 206060_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7905)
PTPN22 | 206060_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6864)
PTPN22 | 206060_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9867)
PTPN22 | 206060_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.411)
PTPN22 | 206060_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PTPN22 | 206060_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4743
p=0.9155
p=0.0595
p=0.1895
p=0.2649
p=0.5834
p=0.3645
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2